Abdomen and pelvis: overview and surface anatomy

CHAPTER 59

GENERAL STRUCTURE AND FUNCTION OF THE ABDOMINOPELVIC CAVITY

Although often considered separately, the abdomen and pelvis form the largest continuous visceral cavity of the body. Together, they provide multiple vital functions, including: housing and protection of the digestive and urinary tracts and of the internal reproductive organs; a conduit for neurovascular communication between the thorax and lower limb; support and attachment for the external genitalia; access to and from the internal reproductive organs and urinary tract; assistance with physiological functions such as respiration, defecation and micturition; support for the vertebral column in weight-bearing, maintenance of posture and movement; and, in women, the ability to support human gestation.

MUSCULOSKELETAL FRAMEWORK OF THE ABDOMEN AND PELVIS

The walls of the abdominopelvic cavity consist of five lumbar vertebrae and their intervening intervertebral discs (lying in the posterior midline); the muscles of the anterior abdominal wall lying anteriorly (rectus abdominis) and anterolaterally (transversus abdominis, internal oblique and external oblique); the muscles of the posterior abdominal wall (psoas, quadratus lumborum and the diaphragm); the bony 'basin' formed by the walls of the false and true pelvis; the muscles of the pelvic floor and perineum lying inferiorly; and the diaphragm lying superiorly (Fig. 59.1). In addition, the upper abdominal cavity gains protection from the lower six ribs and their cartilages, even though these structures are technically part of the thoracic wall.

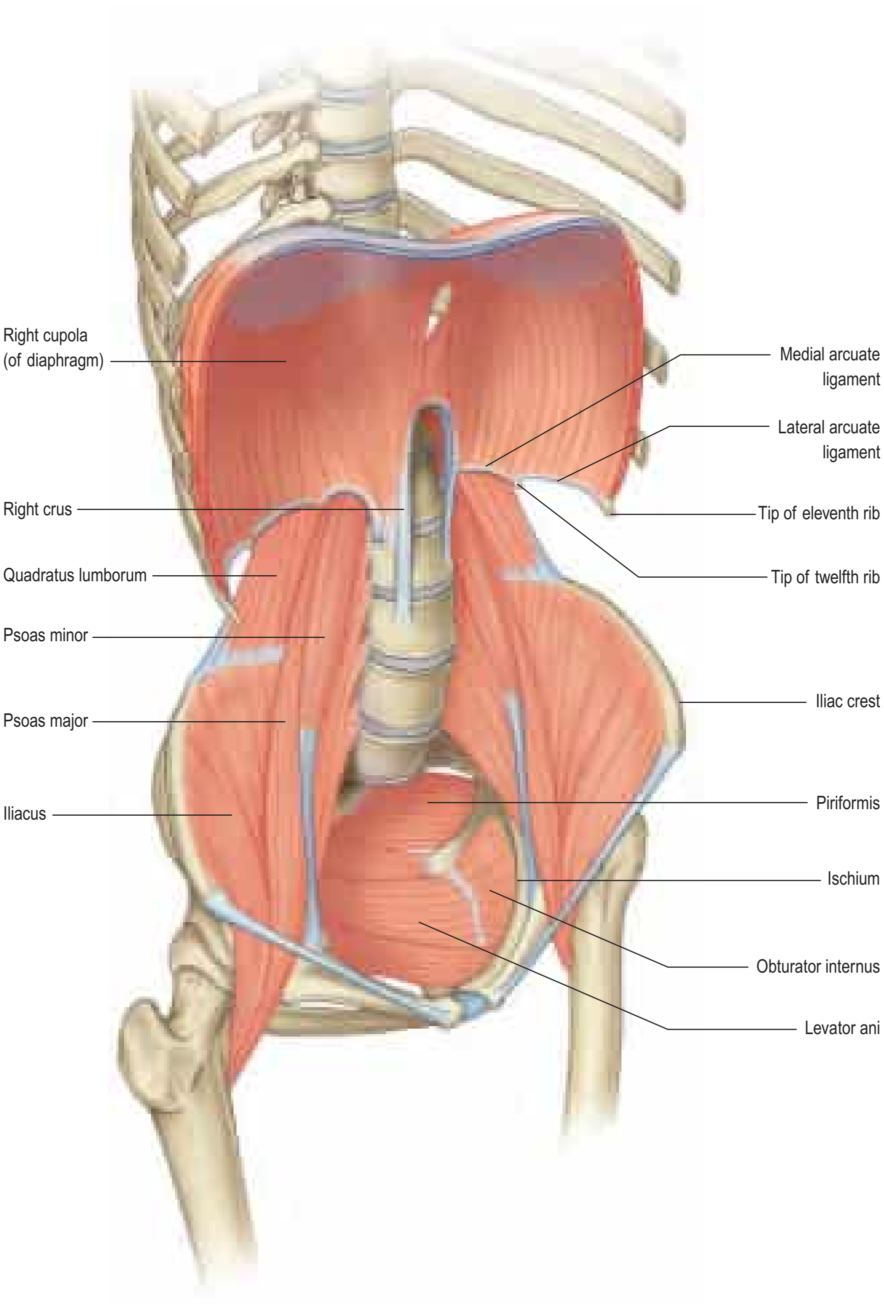

Fig. 59.1 The bony and muscular structures making up the abdominopelvic 'cavity'. The anterolateral abdominal muscles have been removed for clarity.

The muscles of the abdominal wall play an important role in movement of the trunk (flexion, extension and rotation). The anterolateral muscles, in particular, provide assistance with rotation of the thorax in relation to the pelvis (or vice versa if the thorax is fixed).

The abdominal cavity is somewhat kidney-shaped in horizontal cross-section due to the posterior indentation of the vertebral column. Consequently, there are two distinct paravertebral gutters on either side of the spine. The lordosis of the lumbar spine combined with the backward angulation of the sacrum gives each paravertebral gutter a parabola shape in sagittal section. In standing, the pelvic brim lies at an angle of about 55° to the horizontal such that the anterior superior iliac spines and pubic tubercles are in approximately the same coronal plane. The mean angle between the upper border of the first sacral vertebra and the horizontal plane is about 40–45° (Woon et al 2013).

Thoracoabdominal interface

The diaphragm constitutes the interface between the thoracic and abdominal cavities (Ch. 55). Three principal pathways exist between the two cavities across the diaphragm: the caval opening in the central tendon transmits the inferior vena cava and right phrenic nerve; the oesophageal hiatus, encircled by the right crus of the diaphragm, transmits the oesophagus, vagal trunks and vessels; and the aortic hiatus, posterior to the median arcuate ligament of the diaphragm, transmits the aorta, thoracic duct and, usually, the azygos vein. The hemiazygos vein usually enters the thorax through the left crus of the diaphragm. Other lymphatics from the abdomen drain to the thorax alongside the inferior vena cava and via small vessels passing through and around the diaphragm. Thoracic splanchnic nerves reach the abdomen through the diaphragmatic crura and behind the medial arcuate ligaments, and the left phrenic nerve pierces the muscle of the left hemidiaphragm. The subcostal vessels pass into the abdomen beneath the lateral arcuate ligaments of the diaphragm. Anteriorly, the superior epigastric vessels pass between the costal and xiphoid attachments of the diaphragm. Neurovascular structures also cross between the thorax and abdomen within the subcutaneous tissues.

Pelvis–lower limb interface

The pelvis forms an integral part of the bony structure of both the abdominopelvic cavity and the lower limb. It transmits the weight of the upright body, as well as providing a stable platform for movement of the hip joint and bipedal locomotion. Its bony surfaces provide attachment sites for the muscles of the buttock and thigh (Ch. 80), the pelvic floor and perineal membrane, the abdominal wall and lower back. The pelvis also transmits the neurovascular structures that supply the lower limb. There are four principal pathways between the pelvis and lower limb: the interval beneath the inguinal ligament anterior to the superior pubic ramus and ilium, which transmits the femoral neurovascular structures and lymphatics; the greater and lesser sciatic foramina, which transmit the gluteal vessels and nerves, sciatic nerve, and internal pudendal vessels and pudendal nerve; and the obturator foramen, which transmits the obturator nerve, vessels and lymphatics. Autonomic nerves travel with the arterial supply to the lower limb and with the branches of the sacral plexus. Neurovascular structures also cross between the lower limb and pelvis within the subcutaneous tissues.

GENERAL ARRANGEMENT OF ABDOMINOPELVIC AUTONOMIC NERVES

What follows is a brief overview of the autonomic nervous system in the abdominopelvic region; descriptions of the neurovascular supply to individual organs are given in the relevant chapters. The autonomic supply to the abdominal and pelvic viscera is via the abdominopelvic part of the sympathetic chain and the greater, lesser and least splanchnic nerves (sympathetic), and the vagus and pelvic splanchnic nerves (parasympathetic). Numerous interconnections occur between plexuses and ganglia, particularly in the major plexuses around the abdominal aorta; hence, the descriptions tend to be simplifications based on the 'main' supply to each organ (**Fig. 59.2**). The details of the terminations of these fibres are given in the description of the microstructure of the gut wall.

As a general rule, sympathetic neurones from the abdominopelvic autonomic plexuses inhibit visceral smooth muscle motility and glandular secretions, induce sphincter contraction and cause vasoconstriction. Parasympathetic stimulation leads to opposing effects. Visceral afferents also pass through these autonomic plexuses.

Sympathetic innervation

The cell bodies of neurones of the sympathetic supply of the abdomen and pelvis lie in the intermediolateral grey matter of the first to the twelfth thoracic and the first two lumbar spinal segments. Myelinated axons from these neurones travel in the ventral ramus of the spinal nerve of the same segmental level, leaving it via a white ramus communicans to enter a thoracic or lumbar paravertebral sympathetic ganglion. Visceral branches may exit at the same level or ascend or descend several levels in the sympathetic chain before exiting; they leave the ganglia without synapsing and pass medially, giving rise to the paired greater, lesser and least splanchnic nerves, and the lumbar and sacral splanchnic nerves. Axons destined to supply somatic structures synapse in the sympathetic ganglion of the same level, and postganglionic, unmyelinated axons leave the ganglion as one or more grey rami communicantes to enter the spinal nerve of the same segmental level.

Greater splanchnic nerve

On each side, the greater splanchnic nerve is derived from the medial, visceral branches of the fifth to ninth thoracic sympathetic ganglia. It gives off branches to the descending aorta and enters the abdomen through the fibres of the ipsilateral crus of the diaphragm, on which it descends anteroinferiorly. The main trunk of the nerve enters the superior aspect of the coeliac ganglion, where most of the preganglionic fibres synapse (but not those destined for the suprarenal medulla).

Lesser splanchnic nerve

On each side, the lesser splanchnic nerve is derived from the medial, visceral branches of the tenth and eleventh thoracic ganglia (or ninth and tenth). It enters the abdomen running through the lowermost fibres of the ipsilateral crus of the diaphragm or under the medial arcuate ligament, and then lies on the crus as it runs anteroinferiorly. The trunk of the nerve joins the aorticorenal ganglion and may give branches to the lateral aspect of the coeliac ganglion. It occasionally joins the greater and least splanchnic nerves as a single splanchnic nerve.

Least splanchnic nerve

On each side, the least splanchnic nerve is derived from the medial, visceral branches of the eleventh and/or twelfth thoracic ganglia. It enters the abdomen medial to the sympathetic chain under the medial arcuate ligament of the diaphragm and runs inferiorly to join the renal plexus. The trunk of the nerve enters the aorticorenal ganglion and may give branches to the lateral aspect of the coeliac ganglion. It is sometimes part of the lesser splanchnic nerve, when it forms a twig that enters the renal plexus just below the aorticorenal ganglion.

The thoracic splanchnic nerves are subject to considerable individual variation in their origin and distribution (Loukas et al 2010). For example, the greater splanchnic nerve may receive a contribution from the fourth or tenth thoracic sympathetic ganglia or originate only from the sixth to ninth ganglia.

Lumbar sympathetic system

The lumbar part of each sympathetic trunk usually contains four interconnected ganglia lying on the anterolateral aspects of the lumbar vertebrae along the medial margin of psoas major (see Figs 59.2, 62.14) (Murata et al 2003). Superiorly, it is continuous with the thoracic sympathetic trunk posterior to the medial arcuate ligament. Inferiorly, it passes posterior to the common iliac vessels and is continuous with the sacral sympathetic trunk. On the right side, it lies posterior to the inferior vena cava; on the left, it is posterior to the lateral aortic lymph nodes. It is anterior to most of the lumbar vessels but may pass behind some lumbar veins.

On each side, the first, second and, sometimes, the third lumbar ventral spinal rami are connected to the lumbar sympathetic trunk by white rami communicantes. All lumbar ventral rami are joined near their origins by long, slender, grey rami communicantes from the four lumbar sympathetic ganglia. Their arrangement is irregular: one ganglion may give rami to two or three lumbar ventral rami, one lumbar ventral ramus may receive rami from two ganglia, or grey rami may leave the sympathetic trunk between ganglia (Murata et al 2003).

Somatic and vascular branches

Sympathetic nerve branches accompany the lumbar arteries round the sides of the vertebral bodies, medial to the fibrous arches to which psoas major is attached, to provide sympathetic innervation to the lumbar somatomes. Branches from the lumbar ganglia innervate the abdominal

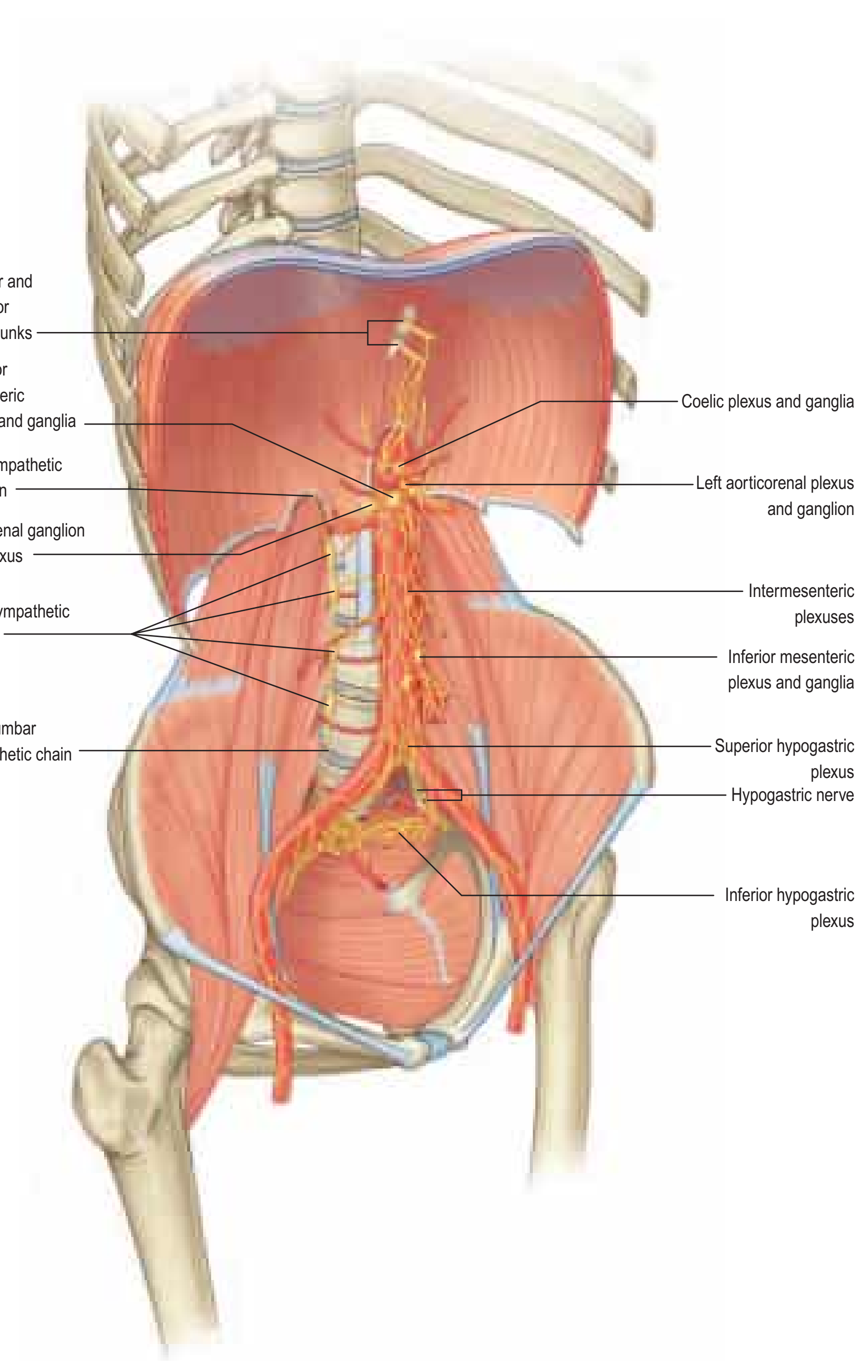

Fig. 59.2 The overall arrangement of the autonomic plexuses of the abdominal and pelvic viscera.

aorta and common, external and internal iliac arteries, forming delicate nerve plexuses that extend along the vessels. Other postganglionic sympathetic nerves to vessels and skin travel with somatic nerves. Thus, the femoral nerve carries vasoconstrictor sympathetic nerves to the femoral artery and its branches in the thigh, as well as sympathetic fibres in its cutaneous branches. Postganglionic fibres travelling with the obturator nerve supply the obturator artery and the skin of the medial thigh. Sympathetic denervation of vessels in the lower limb can be effected by removing or ablating the upper three lumbar ganglia and intervening parts of the sympathetic trunk; this procedure may be useful in treating some varieties of vascular insufficiency of the lower limb.

Lumbar splanchnic nerves

Four lumbar splanchnic nerves pass as medial branches from the ganglia to join the coeliac, inferior mesenteric and superior hypogastric plexuses. The first lumbar splanchnic nerve, from the first ganglion, gives branches to the coeliac, renal and inferior mesenteric plexuses. The second nerve joins the inferior part of the intermesenteric or inferior mesenteric plexus. The third nerve arises from the third or fourth ganglion and joins the superior hypogastric plexus. The fourth lumbar splanchnic nerve from the lowest ganglion passes anterior to the common iliac vessels to join the lower part of the superior hypogastric plexus, or the hypogastric nerves. It is important to note that lumbar splanchnic sympathetic nerves contribute to the superior and inferior hypogastric plexuses and, therefore, contribute to the innervation of the bladder neck, ductus deferens and prostate, among other structures. Damage to these nerves, e.g. during aortoiliac surgery, can result in sexual dysfunction.

Pelvic sympathetic system

The sacral region of the sympathetic trunk usually consists of four or five ganglia located medial or anterior to the anterior sacral foramina beneath the presacral fascia (Oh et al 2004). Occasionally, one or two coccygeal ganglia are present. The sacral sympathetic trunk is continuous above with the lumbar sympathetic trunk, and preganglionic fibres descend from the lower lumbar spinal cord segments via this route. The first sacral sympathetic ganglion is the largest. More caudal ganglia become progressively smaller (Blaszczyk 1981). The sacral sympathetic chain is often asymmetric, with absent or fused ganglia, and cross-communications between each side are frequent. Each ganglion sends at least one grey ramus communicans to its adjacent spinal nerve but up to 11 such branches from a single ganglion have been reported (Potts 1925).

The pelvic sympathetic chain converges caudally to form a solitary retroperitoneal structure, the ganglion impar (or ganglion of Walther), which lies at a variable level between the sacrococcygeal joint and the tip of the coccyx; it is occasionally paired, unilateral or absent (Oh et al 2004). It conveys sympathetic efferents to and nociceptive afferents from the perineum and terminal urogenital regions. Ganglion impar blockade may be used to treat intractable perineal pain of sympathetic origin in patients with pelvic cancers (Toshniwal 2007).

Somatic and vascular branches

Grey rami communicantes containing postganglionic sympathetic nerves pass from the pelvic sympathetic ganglia to the sacral and coccygeal spinal nerves. There are no white rami communicantes at this level. The postganglionic fibres are distributed via the sacral and coccygeal plexuses (Woon and Stringer 2014). Thus, sympathetic fibres in the tibial nerve are conveyed to the popliteal artery and its branches in the leg and foot, whilst those in the pudendal and superior and inferior gluteal nerves accompany these arteries to the perineum and buttocks. Small branches also travel with the median and lateral sacral arteries.

Sacral splanchnic nerves

Sacral splanchnic nerves pass directly from the ganglia to the inferior hypogastric plexus and, from there, to pelvic viscera; they usually arise from the first two sacral sympathetic ganglia.

Parasympathetic innervation

The parasympathetic neurones innervating the abdomen and pelvis lie either in the dorsal motor nucleus of the vagus nerve or in the intermediolateral grey matter of the second, third and fourth sacral spinal segments. The vagus nerves supply parasympathetic innervation to the abdominal viscera as far as the distal transverse colon, i.e. they supply the foregut and midgut. The hindgut is supplied by parasympathetic fibres travelling via the pelvic splanchnic nerves (see below); the overlap between these two supplies is variable. The vagal trunks are derived from the oesophageal plexus and enter the abdomen via the oesophageal hiatus, closely related to the anterior and posterior walls of the abdominal oesophagus (see Fig. 56.6). The anterior vagal trunk is mostly derived from the left vagus and the posterior from the right vagus. The nerves supply the intra-abdominal oesophagus and stomach directly. The anterior trunk gives off a hepatic branch, which innervates the liver parenchyma and vasculature, the biliary tree including the gallbladder, and the structures in the free edge of the lesser omentum. The posterior trunk supplies branches to the coeliac plexus; these fibres frequently constitute the largest portion of the fibres contributing to the plexus. They arise directly from the posterior vagal trunk and from its gastric branch, and run beneath the peritoneum, deep to the posterior wall of the upper part of the lesser sac, to reach the coeliac plexus. Their synaptic relays with postganglionic neurones are situated in the myenteric (Auerbach's) and submucosal (Meissner's) plexuses in the wall of the gut (see below).

Pelvic splanchnic nerves

Pelvic splanchnic nerves to the pelvic viscera travel in the anterior rami of the second, third and fourth sacral spinal nerves. They leave the nerves as they exit the anterior sacral foramina and pass in the presacral tissue as a fine network of branches that are distributed to three principal destinations. Most pass anterolaterally into the network of nerves that form the inferior hypogastric plexus; from here, they pass to the pelvic viscera. Some join directly with the hypogastric nerves and ascend out of the pelvis, as far as the superior hypogastric plexus; from here, they are distributed with branches of the inferior mesenteric artery (p. 1153). A few run superolaterally in the presacral tissue, over the pelvic brim anterior to the left iliac vessels, and pass directly into the tissue of the retroperitoneum and the mesentery of the sigmoid and descending colon.

The pelvic splanchnic nerves are motor to the smooth muscle of the hindgut and bladder wall, supply vasodilator fibres to the erectile tissue of the penis and clitoris, and are secretomotor to the hindgut.

Abdominopelvic autonomic plexuses and ganglia

The abdominopelvic autonomic plexuses are somewhat variable and often fuse or are closely interrelated. The following descriptions recognize their main features (Figs 59.3–59.4).

Coeliac plexus

The coeliac plexus is located at the level of the twelfth thoracic and first lumbar vertebrae, and is the largest major autonomic plexus. It is a dense network that unites the coeliac ganglia, and surrounds the coeliac artery and the root of the superior mesenteric artery. It is posterior to the stomach and lesser sac, and anterior to the crura of the diaphragm

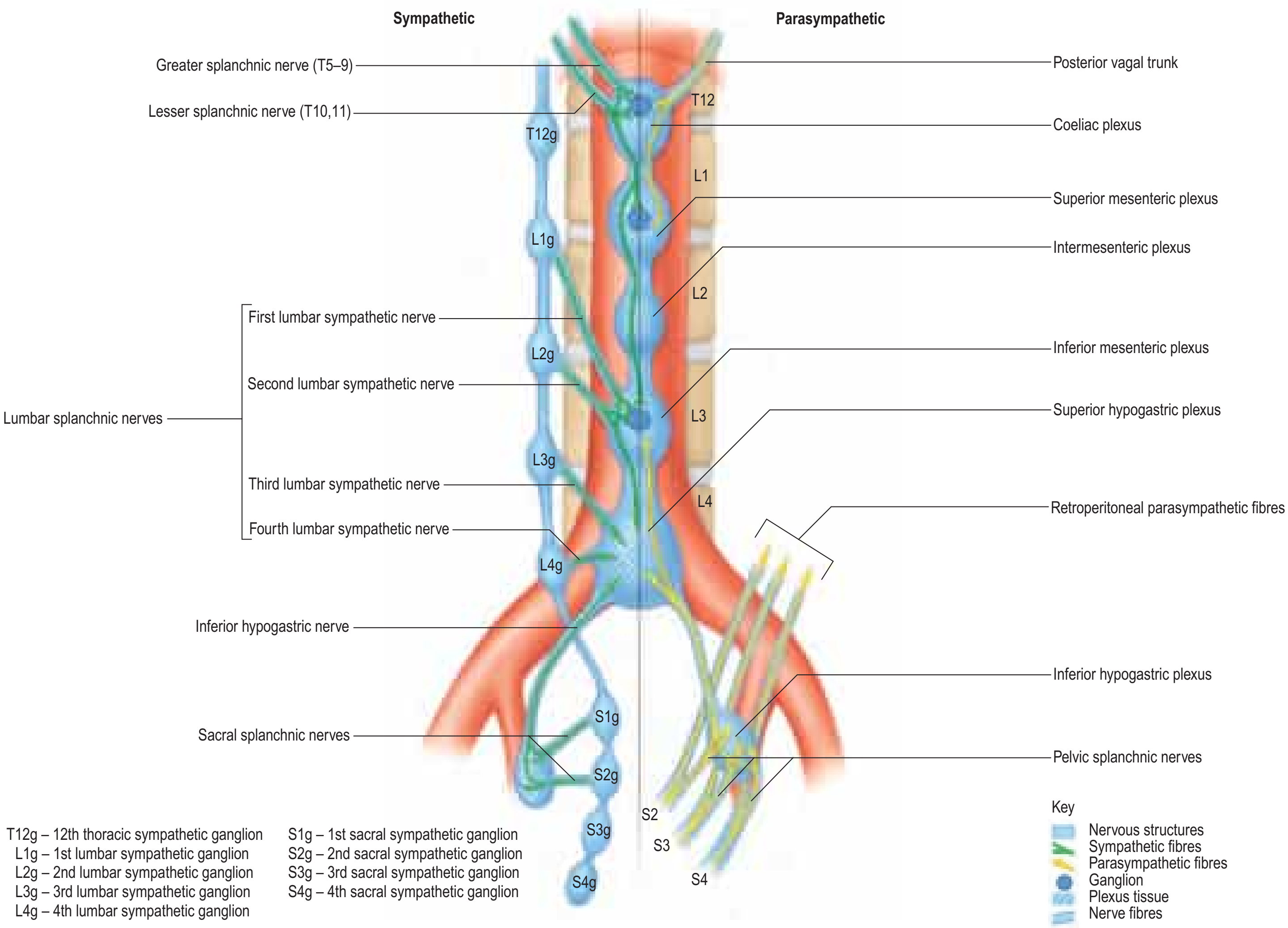

Fig. 59.3 The autonomic plexuses innervating the abdominal and pelvic viscera: schematic diagram.

SECTION 8

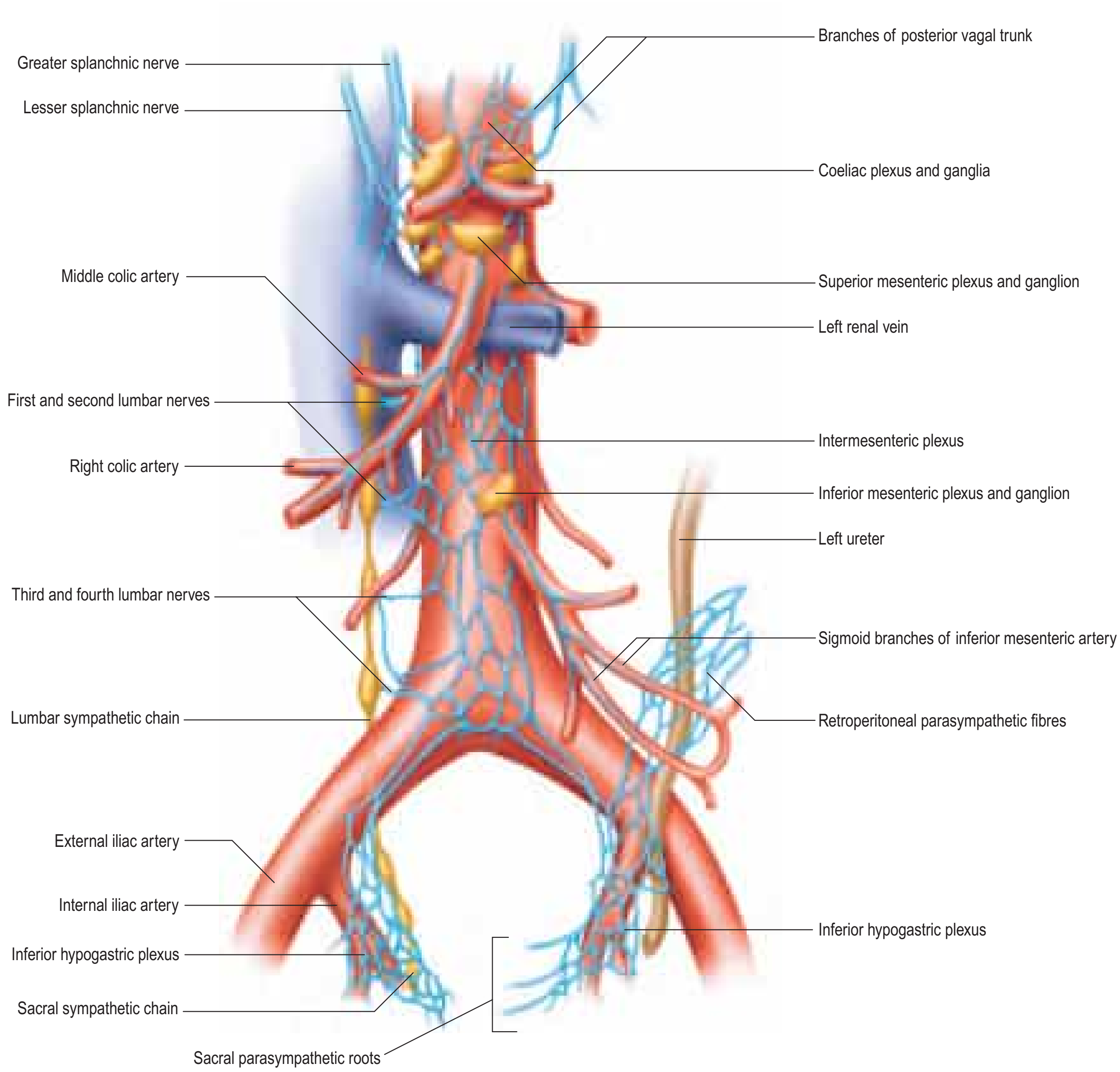

Fig. 59.4 The autonomic plexuses of the abdominal and pelvic viscera.

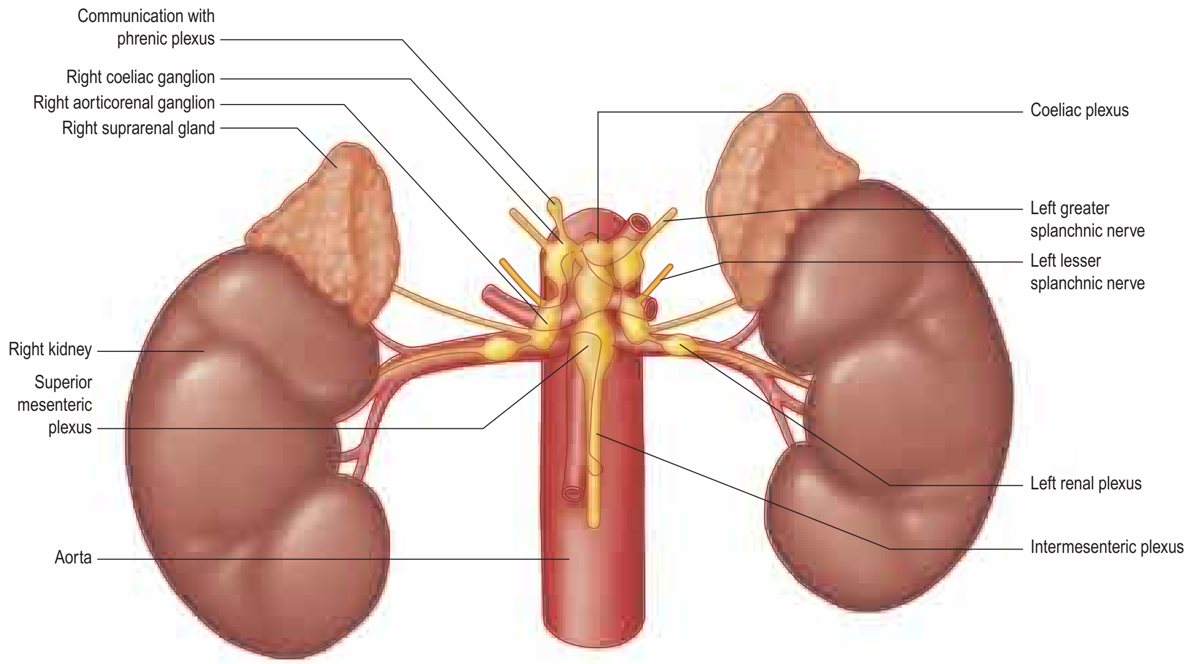

Fig. 59.5 Distribution of the upper abdominal autonomic plexuses.

and the beginning of the abdominal aorta, and lies medial to the suprarenal glands. The plexus and ganglia receive the greater and lesser splanchnic nerves and branches from the vagal trunks. The plexus is in continuity with small branches along adjacent arteries and is connected to the phrenic, splenic, hepatic, superior mesenteric, suprarenal, renal and gonadal plexuses (Fig. 59.5). Visceral afferents in the coeliac plexus convey pain and other sensations from upper abdominal viscera. Anaesthesia or ablation of these nerves (coeliac plexus block) is sometimes undertaken to treat intractable pain from pancreatic disorders.

Coeliac and aorticorenal ganglia

The coeliac ganglia are irregular neural masses, measuring approximately 2 cm across, located between the origin of the coeliac trunk and superior mesenteric artery, medial to the suprarenal glands and anterior to the crura of the diaphragm (Zhang et al 2006). They vary in number, size, shape and precise location; there are often two, one on each side. The right ganglion is frequently posterior to the inferior vena cava, and the left ganglion often lies posterior to the origin of the splenic artery. The ipsilateral greater splanchnic nerve joins the upper part of each ganglion and the lesser splanchnic nerve joins the lower part. The lowermost part of each ganglion forms a distinct subdivision, usually termed the aorticorenal ganglion, which receives the ipsilateral lesser splanchnic nerve and gives origin to the majority of the renal plexus (which, most commonly, lies anterior to the origin of the renal artery).

Phrenic plexus

The phrenic plexus is a periarterial extension of the coeliac plexus around the right inferior phrenic artery, adjacent to the right crus of the diaphragm (Rusu 2006). It contains one or two phrenic ganglia and a definable nerve trunk that connects the coeliac plexus and phrenic nerve. The plexus supplies branches to the suprarenal glands and diaphragm.

Superior mesenteric plexus and ganglion

The superior mesenteric plexus lies in the pre-aortic connective tissue posterior to the pancreas, around the origin of the superior mesenteric artery. It is an inferior continuation of the coeliac plexus, and includes branches from the posterior vagal trunk and coeliac plexus. Its branches accompany the superior mesenteric artery and its divisions. The superior mesenteric ganglion lies superiorly in the plexus, usually above the origin of the superior mesenteric artery.

Intermesenteric plexus

Like other parts of the abdominal aortic autonomic plexus, the intermesenteric plexus is not a discrete structure but is part of a continuous periarterial nerve plexus connected to the gonadal, inferior mesenteric, iliac and superior hypogastric plexuses. It lies on the lateral and anterior aspects of the aorta, between the origins of the superior and inferior mesenteric arteries, and consists of numerous fine, interconnected nerve fibres and a few ganglia continuous superiorly with the superior mesenteric plexus and inferiorly with the superior hypogastric plexus. It is not well characterized but receives parasympathetic and sympathetic branches from the coeliac plexus and additional sympathetic rami from the first and second lumbar splanchnic nerves.

Inferior mesenteric plexus

The inferior mesenteric plexus lies around the origin of the inferior mesenteric artery and is distributed along its branches. It is formed predominantly from the aortic plexus, supplemented by sympathetic fibres from the first and second lumbar splanchnic nerves and ascending pelvic parasympathetic fibres from the inferior hypogastric plexus (via the hypogastric nerves and superior hypogastric plexus). Disruption of the inferior mesenteric plexus alone rarely causes clinically significant disturbances of autonomic function.

Superior hypogastric plexus

The superior hypogastric plexus lies anterior to the aortic bifurcation, the left common iliac vein, median sacral vessels, fifth lumbar vertebral body and sacral promontory, and between the common iliac arteries. It is occasionally referred to as the presacral nerve, but is seldom a single nerve and is prelumbar rather than presacral. It lies within extraperitoneal connective tissue in the midline, extending a little to the left. The breadth of the plexus and its constituent nerves varies; appearances range from a reticular-like arrangement to a band-like structure or one or two distinct nerve trunks (Paraskevas et al 2008). The medial attachment of the sigmoid mesocolon and upper limit of the mesorectum lie anterior to the lower part of the plexus, separated from it by a thin layer of loose connective tissue. The plexus is formed by branches from three main sources: the aortic plexus (sympathetic and parasympathetic), lumbar splanchnic nerves (sympathetic) and pelvic splanchnic nerves (parasympathetic), which ascend from the inferior hypogastric plexus via the right and left hypogastric nerves. Visceral afferents also pass through the plexus. The hypogastric nerves lie in loose connective tissue just posterolateral to the upper mesorectum and pass over the pelvic brim medial to the internal iliac vessels. The superior hypogastric plexus conveys branches to the inferior mesenteric plexus and to the ureteric, gonadal and common iliac nerve plexuses; additional small branches turn abruptly forwards into the upper mesorectum to travel with the superior rectal artery.

Inferior hypogastric plexus

The inferior hypogastric plexus lies in the thin extraperitoneal connective tissue on the pelvic side wall anterolateral to the mesorectum. The internal iliac vessels and the attachments of levator ani and obturator internus lie laterally, and the superior vesical and obliterated umbilical arteries superiorly. In males, the inferior hypogastric plexus lies posterolaterally on either side of the seminal vesicles, prostate and the base of the urinary bladder. In men, the point where the vas (ductus) deferens crosses the ureter provides an approximate guide to the upper limit of the plexus (Mauroy et al 2003). The upper border of the plexus lies beneath the peritoneum of the rectovesical pouch and is in contact with the lateral aspect of the base of the bladder. The anterior border reaches the posterior aspect of the prostate; at its inferior limit, the cavernous (deep, cavernosal) nerve passes forwards to reach the posterolateral aspect of the prostate (see Fig. 75.11). In females, each plexus lies lateral to the uterine cervix, vaginal fornix and the posterior part of the urinary bladder, often extending into the broad ligaments of the uterus. The upper limit of the plexus corresponds approximately to the level where the uterine artery crosses the ureter in the base of the broad ligament (Azaïs et al 2013).

The inferior hypogastric plexus is formed mainly from pelvic splanchnic (parasympathetic) and sacral splanchnic (sympathetic) branches; a smaller contribution is derived from sympathetic fibres (from the lower lumbar ganglia), which descend into the plexus from the superior hypogastric plexus via the hypogastric nerves. It gives origin to a complex network of small pelvic branches, which supply the pelvic viscera either directly or indirectly via periarterial plexuses. The branches of the inferior hypogastric plexus supply the vas deferens, seminal vesicles, prostate, accessory glands and penis in males; the ovary, Fallopian tubes, uterus, uterine cervix and vagina in females; and the urinary bladder and distal ureter in both sexes. The plexus plays a key role in continence and sexual function.

Hypogastric nerves The hypogastric nerves are usually paired nerve bundles but may consist instead of multiple filaments. They contain sympathetic fibres (mostly descending from the superior hypogastric plexus) and parasympathetic fibres (ascending from the inferior hypogastric plexus). The nerves run between the superior and inferior hypogastric plexuses on each side behind the presacral fascia medial to the internal iliac vessels and lateral to the anterior sacral foramina.

Other autonomic plexuses and ganglia

Additional plexuses are described for abdominopelvic viscera, such as the hepatic and gonadal plexuses. Each tends to lie around the main arterial supply to the organ. They receive both sympathetic and parasympathetic fibres from one or more of the major autonomic plexuses and have one or more ganglia. Thus, autonomic ganglia supplying the testis are found around the origin of the testicular arteries from the abdominal aorta and have connections with the renal, lumbar and intermesenteric autonomic plexuses (Motoc et al 2010).

Para-aortic bodies

The para-aortic bodies (also known as paraganglia or, collectively, as the organ of Zuckerkandl) are collections of neural crest-derived chromaffin tissue found in close relation to the aortic autonomic plexuses. They are relatively large in the fetus, reach a maximum size at around 3 years of age, and have usually regressed by adulthood. They are most commonly found as a pair of bodies lying anterolateral to the aorta in the region of the inferior mesenteric and superior hypogastric plexuses, but multiple smaller collections may be present. Occasionally, they are found as high as the coeliac plexus or as low as the inferior hypogastric plexus in the pelvis, or may be closely applied to the sympathetic ganglia of the lumbar chain. Scattered cells that persist into adulthood may, rarely, be the sites of paraganglioma (extra-adrenal phaeochromocytoma) (Subramanian and Maker 2006).

GENERAL ARRANGEMENT OF ABDOMINOPELVIC VASCULAR SUPPLY

The major vessels that occupy the abdomen and pelvis not only supply the viscera, retroperitoneal structures and much of the bony and soft tissue walls of both cavities, but also course through the cavities *en route* to supply the lower limbs. The arteries and systemic veins of the abdomen and pelvis lie predominantly posteriorly in the abdomen and posterolaterally in the pelvis. From the caval and aortic openings in the diaphragm, they follow the general parabolic shape of the lumbar spine. The pelvic divisions follow the contours of the brim and side wall of the true pelvis (**Figs 59.6–59.7**). The individual parts of the aortoiliac and iliocaval systems are described in the relevant chapters.

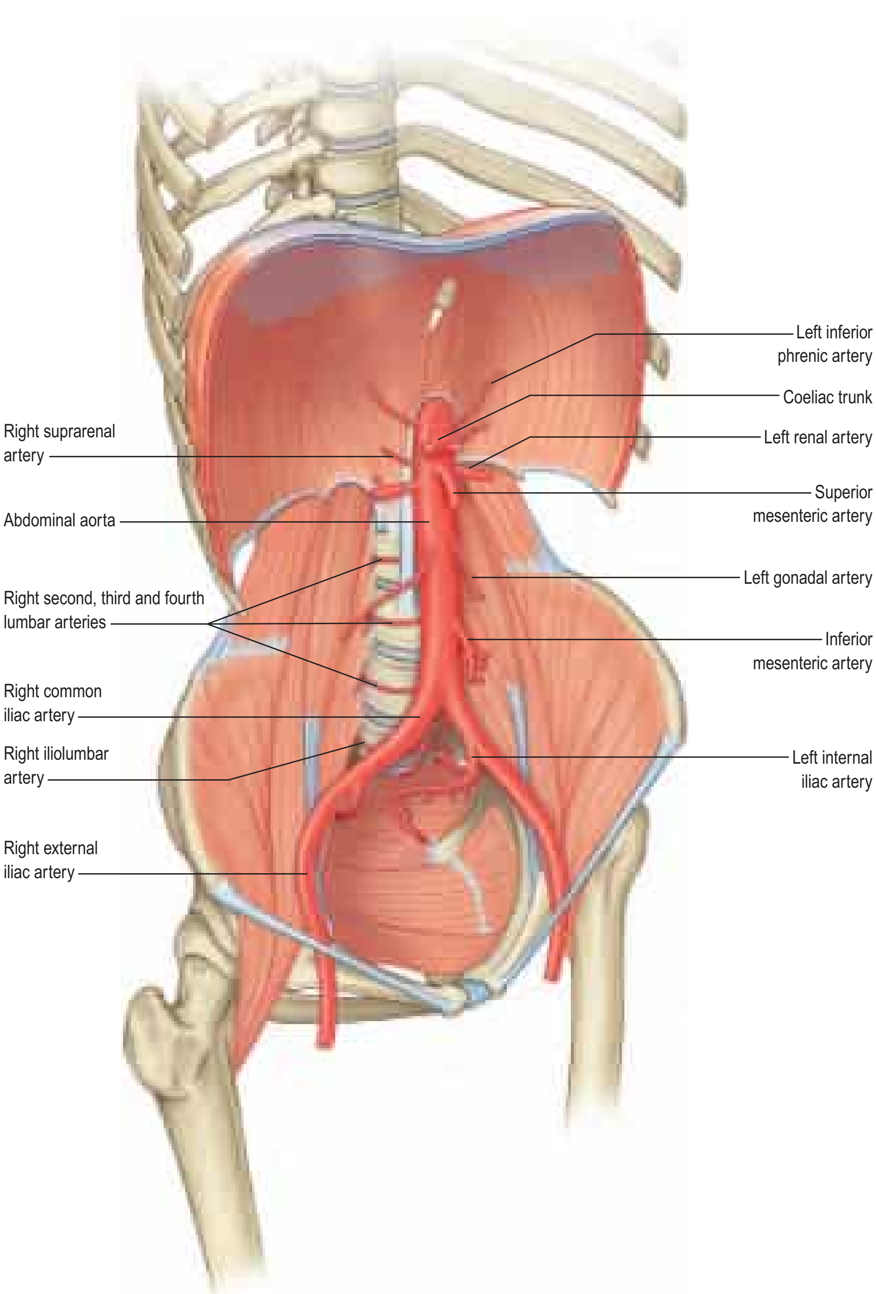

Fig. 59.6 The overall arrangement of the aortoiliac arterial tree of the abdomen and pelvis.

Arterial supply to the gastrointestinal tract

The arterial supply to the gastrointestinal tract is derived from the anterior midline visceral branches of the aorta. There are usually three anterior branches: the coeliac trunk and the superior and inferior mesenteric arteries. Major variants in the origin of the arteries are uncommon (Ch. 62). Accessory or replaced branches to the abdominal viscera are more common. The inferior mesenteric artery almost always arises separately but replaced, accessory or anastomotic vessels may arise from the proximal superior mesenteric artery or its branches. Occasionally, these may contribute to the arterial supply of the inferior mesenteric artery (Horton and Fishman 2002).

The coeliac trunk and its branches supply the part of the foregut that extends from the distal oesophagus to the mid part of the descending duodenum, together with associated viscera (liver, biliary tree, spleen, dorsal pancreas, greater and lesser omenta). The superior mesenteric artery supplies the midgut, which extends from the mid descending duodenum to the distal third of the transverse colon (including jejunum, ileum, caecum, appendix, ascending and transverse colon). The inferior mesenteric artery supplies the part of the hindgut that extends from the distal transverse colon to the upper anal canal. Even in the absence of accessory arteries, numerous anastomoses exist between these vascular territories. For example, anastomoses around the head of the pancreas and the duodenum are formed between the anterior and posterior superior pancreaticoduodenal arteries and the anterior and posterior inferior pancreaticoduodenal arteries, and between the posterior superior pancreaticoduodenal artery and jejunal arteries (Ch. 69). Anastomoses between the territories of the superior and inferior mesenteric arteries are less numerous; the most consistent is the marginal artery of the colon (Ch. 66).

Portal venous system

The (hepatic) portal system, like all portal venous systems, connects two capillary beds: that of the abdominal and pelvic parts of the gut from the abdominal part of the oesophagus to the lower anal canal, together with all derived organs other than the liver (i.e. the pancreas, gallbladder and spleen), and the hepatic sinusoidal 'capillary' bed. The intrahepatic portal vein ramifies and ends in the hepatic sinusoids; from here, blood drains to central veins that converge to form the hepatic veins which drain into the inferior vena cava (Ch. 67). In adults, the portal vein has no valves. In fetal and early postnatal life, valves are demonstrable in tributaries of the portal vein but they usually atrophy with further maturation.

Portal vein

The portal vein is formed behind the neck of the pancreas at the level of the L1/2 intervertebral disc (in the transpyloric plane) from the convergence of the superior mesenteric and splenic veins (Fig. 59.8). It is approximately 8 cm long in the adult and ascends obliquely to the right behind the first part of the duodenum, the common bile duct and gastroduodenal artery; at this point, it is directly anterior to the inferior vena cava. It enters the right border of the lesser omentum and ascends anterior to the epiploic foramen to reach the right end of the porta

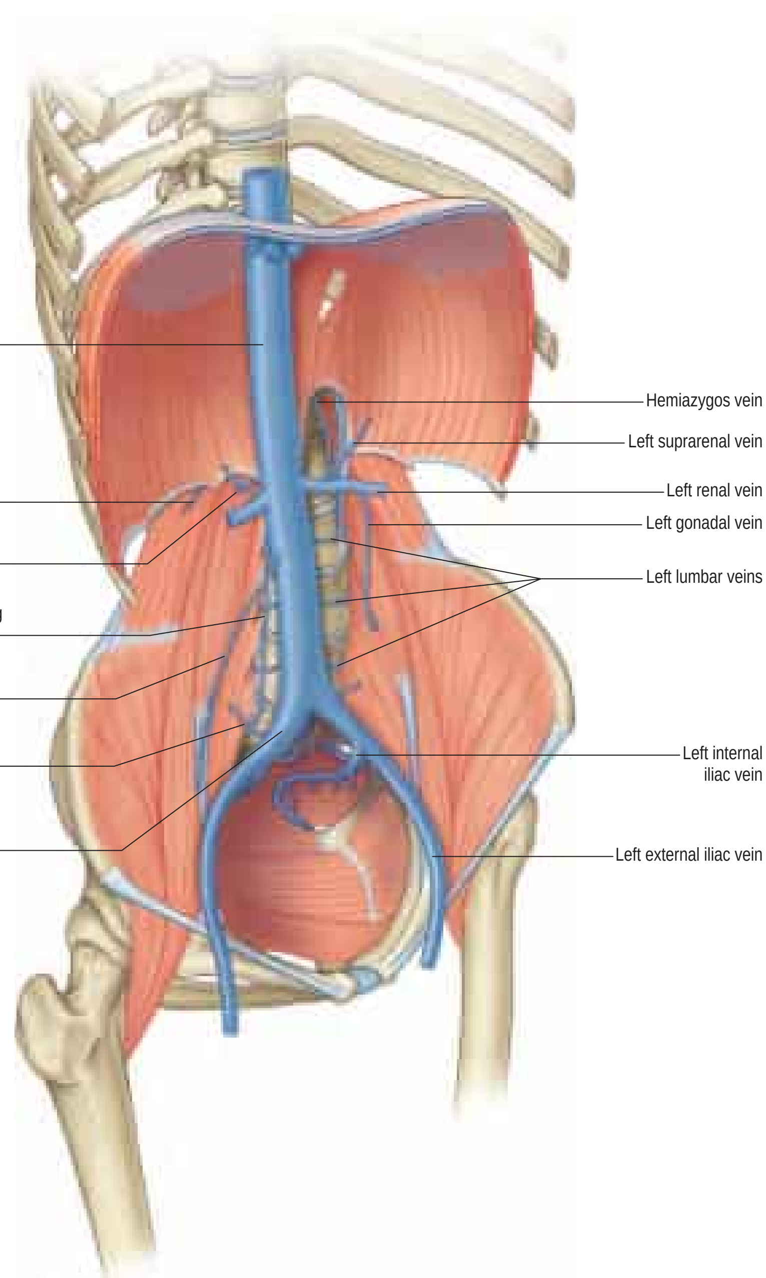

Fig. 59.7 The overall arrangement of the iliocaval venous system of the abdomen and pelvis.

hepatis, where it divides into right and left main branches, which accompany the corresponding branches of the hepatic artery into the liver. In the lesser omentum, the portal vein lies posterior to both the common bile duct and the hepatic artery. It is surrounded by the hepatic nerve plexus and accompanied by lymphatics and some lymph nodes.

Tributaries of the portal vein

The main extrahepatic tributaries of the portal vein are the left gastric (coronary) vein and the posterior superior pancreaticoduodenal vein (Ch. 67). Within the liver, the left branch receives the obliterated umbilical vein via the ligamentum teres, which connects to its vertical portion.

GENERAL MICROSTRUCTURE OF THE GUT WALL

The gut wall displays a common structural plan that is modified regionally to take account of local functional differences. The general microstructure is best appreciated by reference to the development of the gut (Ch. 60). Much of the alimentary canal originates as a tube of endoderm enclosed in splanchnopleuric mesoderm. Its external surface faces the embryonic coelom, and the endodermal lining forms the epithelium of the canal and also the secretory and ductal cells of various glands that secrete into the lumen, including the pancreas and liver. The splanchnopleuric mesoderm forms the connective tissue, muscle layers, blood vessels and lymphatics of the wall, and its external surface becomes the visceral mesothelium or serosa. There is no serosa surrounding the cervical and thoracic portions of the gut, or below where the hindgut traverses the pelvic floor; in these sites, the gut tube is surrounded by a connective tissue adventitia. Neural elements invade the gut from neural crest tissue (Ch. 17). The smooth muscle of the muscularis externa layers of the alimentary canal is supplemented with striated muscle at the beginning (from the branchial arches) and end of the gut tube.

The mature gut wall has four main layers: mucosa, submucosa, muscularis externa and serosa (Fig. 59.9). The mucosa (mucous membrane) is the innermost layer and is subdivided into a lining epithelium, an underlying lamina propria (a layer of loose connective tissue, where many of the glands are also found) and a thin layer of smooth muscle, the muscularis mucosae. The submucosa is a strong and highly vascularized layer of connective tissue. The muscularis externa consists of inner circular and outer longitudinal layers of smooth muscle; an incomplete oblique muscle layer is present only in the stomach. The external surface is bounded by a serosa or adventitia, depending on its position within the body.

Mucosa

Epithelium

The epithelium is the site of secretion and absorption, and provides a defence against various threats, including microorganisms. Its protective function against mechanical, thermal and chemical injury is particularly evident in the oesophagus and anal canal, where it is thick, stratified and covered in mucus, which serves as a protective lubricant. At other sites, the epithelium lining the gut wall is single-layered, and

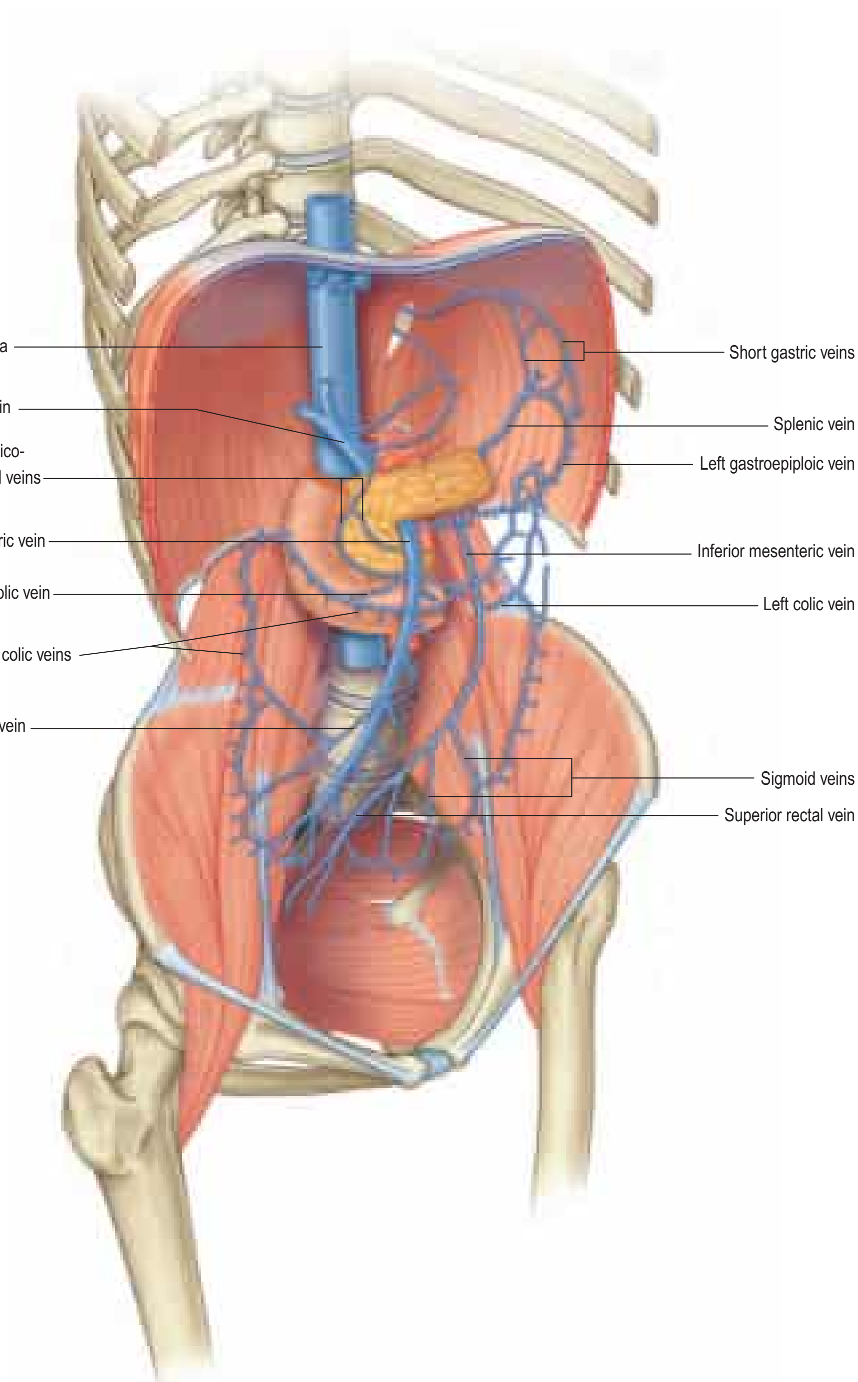

Fig. 59.8 The overall arrangement of the portal venous system draining the abdominal viscera.

either cuboidal (in glands) or columnar. It contains cells modified for absorption, as well as various types of secretory cell.

The barrier function and selectivity of absorption depend on tight junctions over the entire epithelium. The surface area of the lumen available for secretion or absorption is increased by the presence of mucosal folds, crypts, villi and glands (see Fig. 59.9). Microvilli on the surfaces of individual absorptive cells amplify the area of apical plasma membrane in contact with the contents of the gut. Some glands lie in the lamina propria and some in the submucosa; others (the liver and pancreas) are totally external to the wall of the gut. All of these glands drain into the lumen of the gut through individual ducts. The epithelium also contains scattered neuroendocrine (enteroendocrine) cells.

Lamina propria

The lamina propria consists of compact connective tissue, often rich in elastin fibres, which supports the surface epithelium and contains nutrient vessels and lymphatics. Lymphoid follicles are present in many regions of the gut, most notably in Peyer's patches. Cells within the lamina propria are the source of growth factors that regulate cell turnover, differentiation and repair in the overlying epithelium.

Muscularis mucosae

The muscularis mucosae is particularly well developed in the oesophagus and in the large intestine, especially in the terminal part of the rectum. In addition, single muscle cells originating from the muscularis mucosae are found inside the villi or between the tubular glands of the stomach and large intestine. By its contraction, the muscularis mucosae can alter the surface configuration of the mucosa locally, allowing it to adapt to the shapes and mechanical forces imposed by the contents of the lumen, and in the intestinal villi, promoting vascular exchange and lymphatic drainage.

Submucosa

The submucosa contains large bundles of collagen and is the strongest layer of the gut wall. However, it is also pliable and deformable, and can therefore adjust to changes in the length and diameter of the gut. Its contained arterial network is relatively dense and supplies both the mucosa and the muscle coat. The submucosa extends into the rugae of the gastric wall, the plicae circulares of the small intestine (but not the villi), and the folds that project into the lumen of the colon and rectum.

Muscularis externa

The muscularis externa usually consists of distinct inner circular and outer longitudinal layers that create waves of peristalsis responsible for the movement of ingested material through the lumen of the gut. In the stomach, where movements are more complex, there is an incomplete oblique layer of muscle, internal to the other two layers. The

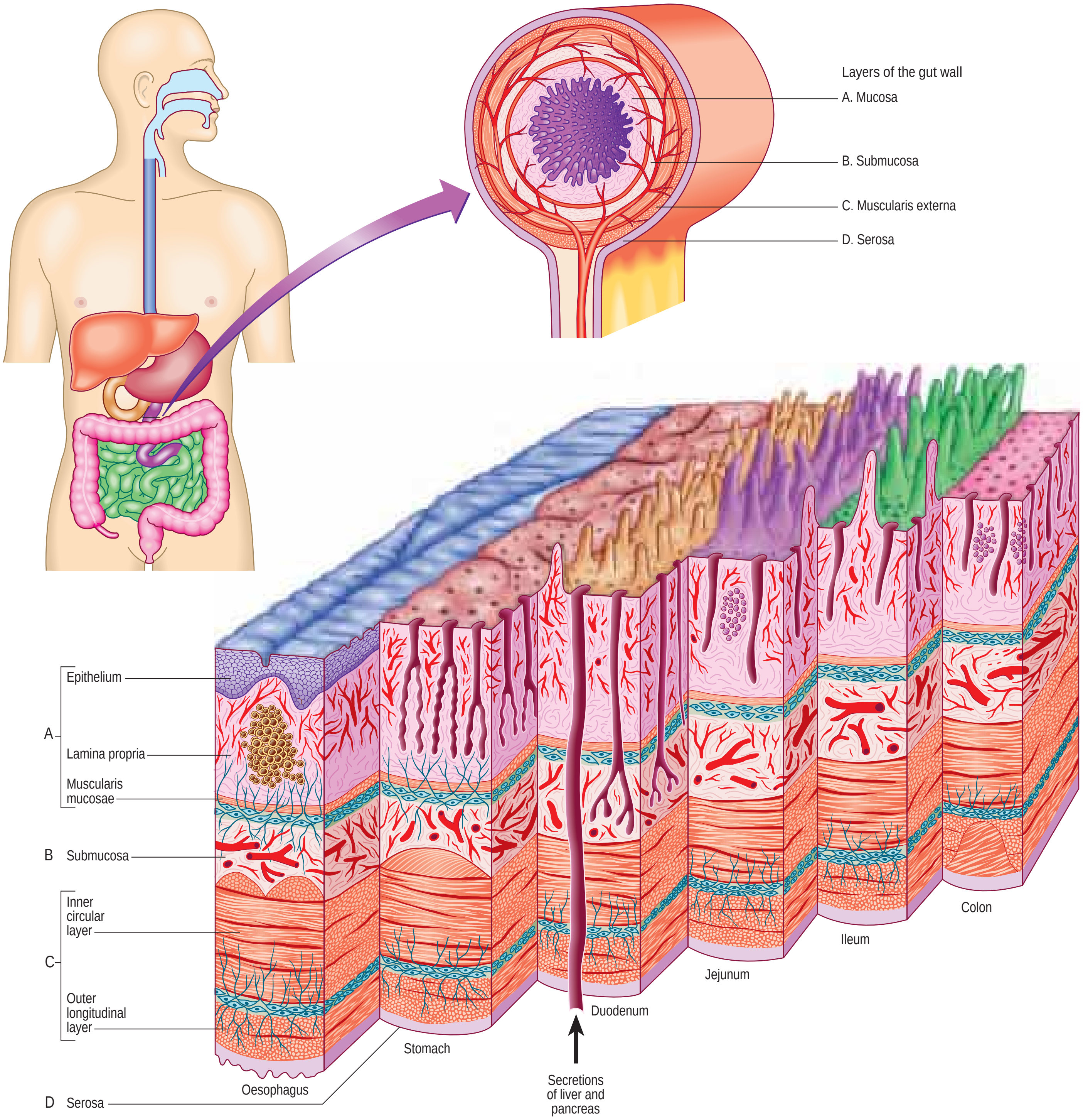

Fig. 59.9 The general arrangement of the alimentary canal to show the layers of the gut wall at the levels indicated.

circular muscle layer is invariably thicker than the longitudinal muscle, except in the colon, where the longitudinal muscle is condensed into three cords (taenia coli).

The muscularis externa is composed almost exclusively of smooth muscle, except in the upper oesophagus, where smooth muscle blends with striated muscle. Although the upper oesophageal musculature resembles that of the pharynx, it is entirely under involuntary control. For most of its length, the smooth muscle of the gut wall consists of ill-defined bundles of cells, typically visceral in type, and somewhat larger than vascular smooth muscle cells. They are approximately 500 μm long, regardless of body size, and are electrically and mechanically coupled. Their fasciculi lack a perimysium but have sharp boundaries.

The arrangement of the musculature allows a segment of gut to undergo extensive changes in diameter (to virtual occlusion of the lumen) and length, although elongation is limited by the presence of mesenteries. The coordinated activity of the two muscle layers produces a characteristic motor behaviour that is mainly propulsive and directed aborally (peristalsis), combined with a non-propulsive motor activity that either mixes the luminal contents, as occurs in the stomach, or partitions them, as occurs at the pyloric sphincter. The muscle maintains a constant volume, so that shortening of a segment of the gut wall is accompanied by an increase in thickness of the muscle layer.

Intestinal smooth muscle exhibits variable and changing degrees of contraction on which rhythmic (or phasic) contractions are superimposed. Slow waves of rhythmic electrical activity, driven by changes in membrane potentials in pacemaker cells (interstitial cells), spread throughout the thickness of the circular and longitudinal smooth muscle coats. After spreading circumferentially, slow waves can move in either oral or anal directions, causing segmental contraction. The distances of propagation and the patterns of this spontaneous activity vary between areas of the intestine. Neural regulation of slow and phasic contractions involves excitatory and inhibitory transmitters that are released from the myenteric plexus. This motor control is closely coordinated with mucosal absorption and secretion, and is mediated via intrinsic nerves in the submucous plexus. The peristaltic reflex occurs during passage of luminal contents down the intestine. It involves ascending contraction and descending relaxation; the sensory limb is mediated by sensory neurones that respond to either mucosal stimulation (intrinsic primary afferents) or muscle stretch (extrinsic afferents).

Interstitial cells

Interstitial cells of Cajal (ICCs) are found throughout the entire length of the gastrointestinal tract, where they lie in close contact with nerve terminals; they have numerous gap junctions with each other and with smooth muscle cells. Distinct networks are found in the myenteric plexus between the circular and longitudinal muscle. Looser arrangements exist within the individual muscle layers and the submucosa of the gut, and isolated or small groups of ICCs are also found in the subserosal region. ICCs originate from mesenchymal cells and resemble smooth muscle cells but have fewer contractile elements and their intermediate filaments contain vimentin rather than desmin. They are involved in the generation of pacemaker signals, the propagation of electrical slow wave activity, neuromuscular transmission, and mechanosensation (p. 1135) (Sanders et al 2014). Defective ICC function has been implicated in a wide range of gastrointestinal motility disorders (Al-Shboul 2013), and in the development of gastrointestinal stromal tumours (Roggin and Posner 2012).

Cells with a similar morphology have been found in association with smooth muscle at many other sites, such as the urethra, ductus deferens, prostate, bladder, corpus cavernosum, ureter, uterine tube, and uterus. They may act as electrical pacemakers in the urethra and ureter but their role at other sites is less certain (Drumm et al 2014).

Serosa and adventitia

A layer of connective tissue, of variable thickness, lies external to the muscularis externa. In many places, it contains adipose tissue. Where the gut is covered by visceral peritoneum, the external layer is a serosa, consisting of mesothelium overlying a thin layer of connective tissue. In extraperitoneal regions, the surfaces of the gut in contact with the peritoneum are covered by serosa, while other parts are covered by connective tissue that blends with the surrounding fasciae and is referred to as an adventitia.

Vascular plexuses

Vascular plexuses are present at various levels of the wall, especially in the submucosa and mucosa; they connect with vessels that supply the surrounding tissues or those entering through the mesentery, and accompany the ducts of outlying glands.

Innervation

The gut wall is densely innervated by both the enteric nervous system (intrinsic control) and the autonomic nervous system (extrinsic control). The latter is not essential for function, as evidenced by intestinal activity after transplantation of the extrinsically denervated gut. In contrast, neuropathies affecting the enteric nervous system, such as Hirschsprung's disease, are potentially life-threatening. For a detailed description of the enteric nervous system, see Furness et al (2014).

Extrinsic innervation

Extrinsic innervation of the gut is from sympathetic, parasympathetic and visceral sensory nerves. The cell bodies of preganglionic parasympathetic efferent axons are found in the vagal dorsal motor nucleus in the medulla oblongata and in the sacral segments of the spinal cord. Their main target in the gut is the enteric neurones of the myenteric plexus. Vagal efferents play a major role in oesophageal propulsion, gastric acid secretion and emptying, gallbladder contraction and pancreatic exocrine secretion. Pelvic efferents innervate the distal colon and rectum. Sympathetic efferent neurones have their cell bodies in the intermediolateral grey matter (lamina VII) of the thoracolumbar segments of the spinal cord; those destined for the smooth muscle of the gut mostly pass through the sympathetic chain without synapsing and relay in prevertebral ganglia (coeliac, mesenteric and pelvic), whereas those mediating vasoconstriction synapse in the sympathetic chain or prevertebral ganglia. Postganglionic sympathetic fibres are distributed with the branches of the coeliac trunk and mesenteric arteries to three principal targets: the myenteric and submucosal ganglia (causing inhibition), blood vessels (inducing vasoconstriction) and sphincter muscle (inducing contraction).

Visceral sensory endings are distributed throughout the layers of the gut wall. They respond to various stimuli, including excessive muscular contraction or distension, ischaemia and inflammation; their cell bodies are located in the nodose ganglion of the vagus nerve (vagal afferents) and in thoracic and lumbosacral dorsal root ganglia (visceral afferents, which run with sympathetic efferents). Their central projections reach the brainstem and spinal cord, respectively. Vagal afferents are more numerous in the foregut than the midgut and are concerned with physiological responses such as satiety, whereas pain and discomfort are mediated by spinal pathways.

Intrinsic innervation

The intrinsic innervation of the gut is derived from enteric neurones distributed throughout its wall in thousands of small ganglia, each consisting of neuronal cell bodies supported by enteric glial cells; it is estimated that there are between 200 and 600 million neurones in the human enteric nervous system. Enteric neurones are derived from neural crest cells. The enteric nervous system innervates the gut from oesophagus to anus, together with the pancreas and biliary tree.

The majority of enteric neurones in the wall of the gut are found in two ganglionated plexuses (Fig. 59.10). The most extensive of these is the myenteric (Auerbach's) plexus, which consists of a network of nerve

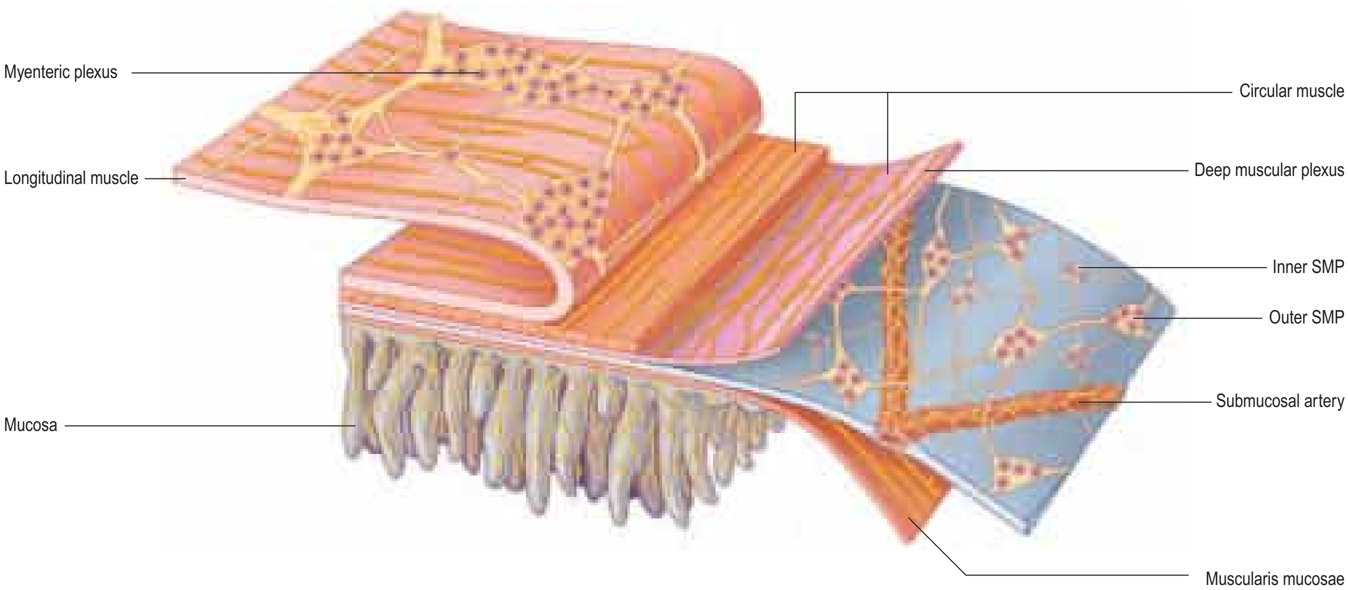

Fig. 59.10 The organization of the enteric nervous system in the human small intestine. There are two ganglionated plexuses: the myenteric plexus between the longitudinal and circular layers of the external musculature, and the submucosal plexus (SMP), which has outer and inner components. Nerve fibre bundles connect the ganglia and form plexuses innervating the longitudinal muscle, circular muscle, muscularis mucosae, intrinsic arteries and the mucosa. Axons of extrinsic origin also run in these nerve fibre bundles. There are also innervations of gastroenteropancreatic (GEP) endocrine cells and gut-associated lymphoid tissue (GALT), which are not shown here. (Redrawn with permission from Furness JB. The enteric nervous system and neurogastroenterology. Nat Rev Gastroenterol Hepatol 2012;9:286–294. Reprinted with permission from Nature Publishing Group.)

fibres and small ganglia lying between the circular and longitudinal layers of the muscularis externa. It runs in continuity from the oesophagus to the anus. The submucosal (Meissner's) plexus has outer and inner components, and is present throughout the intestine but absent or minimal in the oesophagus and stomach.

Interconnecting, non-ganglionated nerve plexuses lie at various levels in the wall of the gut, including the lamina propria (mucosal plexus), at the interface between the submucosa and muscularis externa, between the circular and longitudinal muscles (the non-ganglionated part of the myenteric plexus), and within the serosa.

Individual enteric neurones function in one of several ways: as afferent (sensory) nerves responding to mechanical and chemical stimuli; as efferent (motor) nerves innervating epithelial cells (influencing absorption, secretion and/or the release of hormones), smooth muscle (excitatory or inhibitory), arterioles (vasoconstriction or vasodilation) and lymphoid tissue; or as interneurones that relay and integrate signals. Collectively, they are involved in a hierarchy of enteric reflexes that include local reflexes within the gut wall; reflexes mediated through prevertebral sympathetic ganglia; and reflexes mediated through the central nervous system (the gut–brain axis). The functions of enteric neurones may be further modulated by local environmental factors, e.g. enteric glia, gut microbiota and feeding state, and by general factors in the host, e.g. the immune system, stress and disease (Brierley and Linden 2014). Pathological states become manifest by abnormal secretion, impaired absorption, disordered gastrointestinal motility and pain.

SURFACE ANATOMY OF THE ABDOMEN AND PELVIS

ABDOMINAL PLANES AND REGIONS

For descriptive purposes, the abdomen can be divided into regions using a combination of horizontal and vertical planes that are based on skeletal landmarks. Published descriptions of surface anatomy largely apply to adults, in whom attempts have been made to validate or update standard descriptions using cross-sectional imaging. The extent to which many of these surface projections can be extrapolated to children is uncertain; some differences are inevitable, particularly in infants, given their different visceral and body proportions (Stringer 2011). In this chapter, descriptions of the surface projections of abdominal structures refer to the most commonly encountered arrangement seen in adults. However, it is important to note that there is considerable individual variation in surface anatomy, compounded further by potential variations related to age, sex, posture, respiration, body mass and ethnicity (Mirjalili and Stringer 2012).

Vertical lines and planes

The midline (or median plane) passes through the xiphoid process and the pubic symphysis. There are two paramedian lines, left and right, that extend from the mid-clavicular point to the mid-inguinal point, a point midway between the anterior superior iliac spine and the pubic symphysis (**Fig. 59.11A**); this line crosses the costal margin just lateral to the tip of the ninth costal cartilage. In the upper abdomen, it approximates to the lateral border of rectus abdominis.

Horizontal lines and planes

Several horizontal reference planes have been defined but, with modern cross-sectional imaging, their clinical utility for positioning abdominal viscera has become limited. Nevertheless, they help to conceptualize the relative positions of abdominal viscera with respect to vertebral levels and bony surface landmarks.

The xiphisternal joint most often lies in the same horizontal (axial) plane as the T9 vertebra (Mirjalili et al 2012b).

The transpyloric plane lies midway between the suprasternal notch of the manubrium and the upper border of the pubic symphysis. This corresponds to a plane that is approximately midway between the xiphisternal joint and the umbilicus, or slightly below (Mirjalili et al 2012a). Posteriorly, the plane intersects the lower half of the body of the first lumbar vertebra, the L1/2 intervertebral disc, or the upper half of the body of the second lumbar vertebra in most individuals (range lower T12 to lower L2); it sits higher in males (lower L1–L1/2) than females (lower L2) (Mirjalili et al 2012a). Anteriorly, it intersects the costal margin at the ninth costal cartilage, where the linea semilunaris crosses, and where a distinct 'step' may be palpable in the rib margin. The following structures lie approximately within the transpyloric plane (Mirjalili et al 2012a, 2012b): the origin of the superior mesenteric artery; the origin of the portal vein (from the confluence of the superior mesenteric and splenic veins behind the neck of the pancreas); the hilum of the left kidney (the hilum of the right kidney is slightly lower); the origin of the renal arteries; the level of the duodenojejunal flexure (Mirjalili 2012b); and the termination of the spinal cord. The pylorus may be found in the transpyloric plane but is not a constant feature.

The subcostal plane is defined by a line that joins the lowest point of the costal margin on each side (usually, the tenth costal cartilage); it is most commonly at the level of the second lumbar vertebra (range T12/L1 disc to upper L3) (Mirjalili et al 2012a).

The supracristal plane joins the highest point of the iliac crest on each side. It serves as a useful landmark when performing a lumbar puncture or spinal injection, and usually lies at the level of the L4 body or L4/5 disc and its spinous process (see Fig. 78.16). The bifurcation of the abdominal aorta lies approximately in this plane (Mirjalili et al 2012a).

The transtubercular plane joins the tubercles of the iliac crests and is traditionally reported to lie at the level of the body of L5, near its upper border. It is an approximate landmark for the origin of the inferior vena cava from the confluence of the common iliac veins.

The plane of the pubic crest is in line with the upper border of the pubic symphysis. In supine individuals, this plane frequently intersects the tip of the coccyx and the femoral head (Mirjalili et al 2012a), but there are variations relating to the degree of lumbar lordosis, sacral inclination and pelvic tilt.

Abdominal regions

The abdomen can be divided into nine regions using the subcostal, transtubercular and two paramedian planes (**Fig. 59.11A**). These regions are used in clinical practice for descriptive localization of a patient's pain or tenderness, or the position of a mass, and can be used to reference the position of abdominal viscera. From superior to inferior, the nine regions are: the epigastrium, flanked by the right and left hypochondrium; the umbilical, or central, flanked by the right and left lumbar; and the suprapubic or hypogastrium, flanked by the right and left iliac fossae. In practice, these regions are often loosely defined without strict reference to their boundaries. An alternative system of description involves dividing the abdomen into quadrants using the median plane and a horizontal line passing through the umbilicus.

ANTERIOR ABDOMINAL WALL

Skeletal landmarks

The superior boundary of the anterior abdominal wall is formed by several clear landmarks (**Fig. 59.11B**). The xiphoid process is palpable at the inferior sternum, in the midline. From here, the costal margins can be felt passing inferolaterally from the seventh costal cartilage at the xiphisternal joint to the tip of the twelfth rib (the latter is often difficult to feel in the obese or if it is short). The lower border of the ninth costal cartilage can usually be defined as a distinct 'step' along the costal margin. The lowest part of the costal margin lies in the mid-axillary line and is formed by the inferior margin of the tenth costal cartilage.

The inferior boundary of the anterior abdominal wall is formed, from lateral to medial, by the iliac crest, which descends to the anterior superior iliac spine; the inguinal ligament, which runs obliquely downwards to the pubic tubercle; and the pubic crest, which extends from the pubic tubercle laterally to the pubic symphysis in the midline. The pubic tubercle is palpable on the anterosuperior surface of the pubic bone approximately 2 cm lateral to the midline. The prominent tendon of adductor longus (best felt with the hip flexed, abducted and externally rotated) attaches to the pubis directly below the pubic tubercle, and can therefore be used to verify its position.

The posterolateral boundary of the anterior abdominal wall is the mid-axillary line.

Soft tissue landmarks

Umbilicus

The umbilicus is an obvious but inconstant landmark. In the supine adult, it usually lies around the L4 level (range L2/3 disc to upper S1) (Mirjalili et al 2012b). The umbilicus may lie at a lower level in the erect position, and in children, the obese and in individuals with a pendulous abdomen.

Rectus abdominis

In a thin, muscular individual, the tendinous intersections of rectus abdominis may be visible, especially when the muscle is tensed by lifting the head against resistance or by sitting up; the intersections are usually situated at the level of the umbilicus, the level of the xiphoid process and midway between these two points (**Fig. 59.11B**).

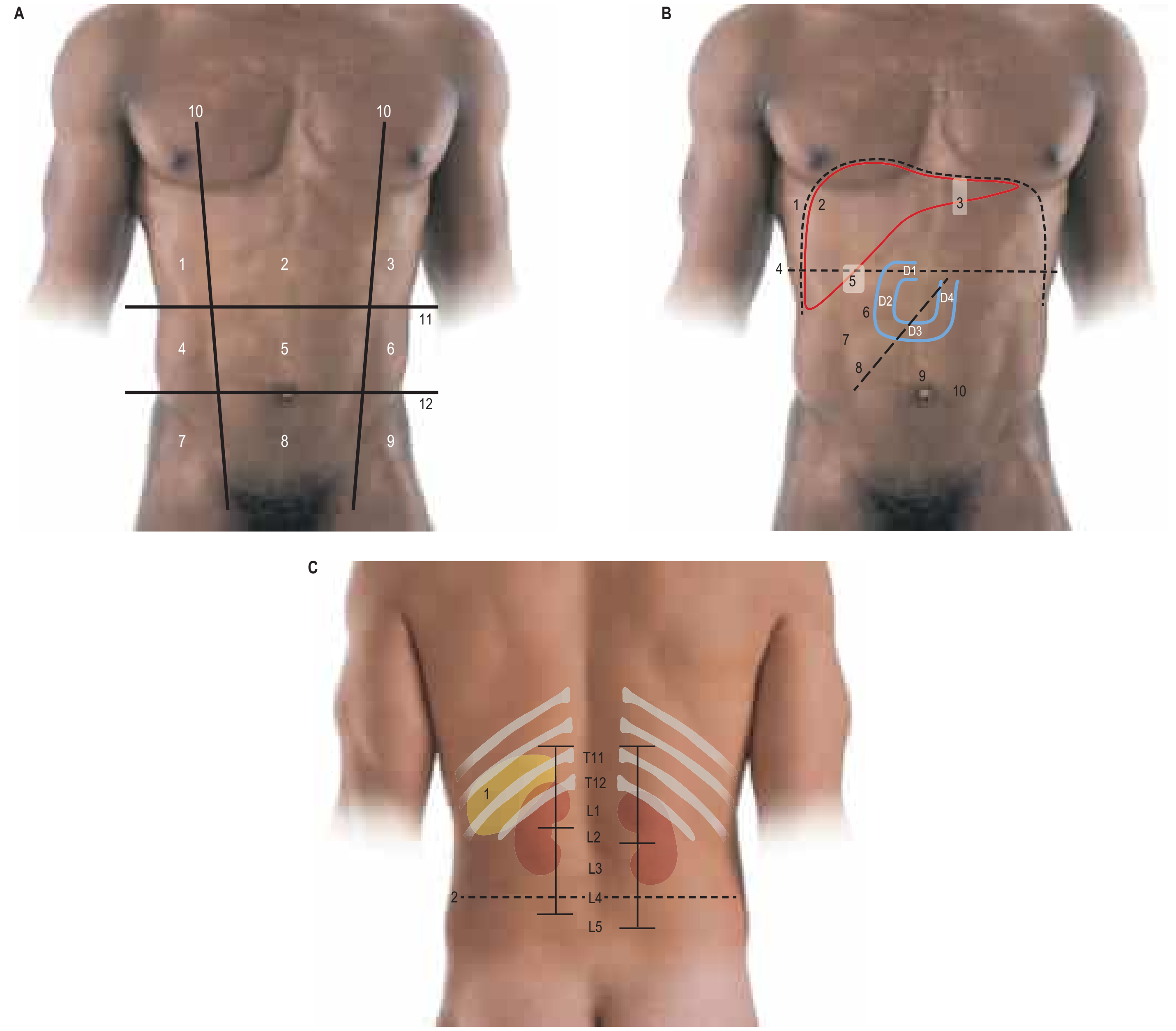

Fig. 59.11 **A**, Nine regions of the anterior abdominal wall. Key: 1, right hypochondrium; 2, epigastrium; 3, left hypochondrium; 4, right lumbar; 5, umbilical/central; 6, left lumbar; 7, right iliac fossa; 8, suprapubic/hypogastrium; 9, left iliac fossa; 10, paramedian line; 11, subcostal plane; 12, transtubercular plane. **B**, The surface projection of the abdominal viscera. Key: 1, diaphragm position: right dome level with fifth intercostal space and left dome with sixth rib; 2, liver: mapped between three points: right fifth rib/intercostal space mid-clavicular line, left fifth intercostal space/sixth rib mid-clavicular line and right tenth costal cartilage mid-axillary line; 3, zone of gastro-oesophageal junction position (white): mainly located posterior to left seventh costal cartilage, at approximately T11; 4, transpyloric plane; 5, zone of gallbladder fundus position (white); 6, duodenum: four parts marked D1–D4; 7, linea semilunaris; 8, position of small intestine mesentery; 9, linea alba; 10, tendinous intersection of rectus abdominis. **C**, The surface position of the spleen and kidneys. Key: 1, spleen; sits deep to ribs 10–12 with long axis aligned with rib 11; 2, supracristal plane. (C, Derived with permission from Mirjalili SA, McFadden SL, Buckenham T, Stringer MD. 2012b A reappraisal of adult abdominal surface anatomy. Clin Anat 25:844–50.)

Linea alba

The linea alba is usually only visible in thin, muscular individuals. It is wider and more obvious above the umbilicus, and is almost linear and less visible below this level (Ch. 61) (Fig. 59.11B; see Fig. 61.2).

Linea semilunaris

The linea semilunaris lies along the lateral margin of the rectus sheath and is visible as a shallow, curved groove in muscular individuals, particularly when the abdominal muscles are tensed, e.g. by sitting up from the lying position (Fig. 59.11B). It passes from the ninth costal cartilage to the pubic tubercle.

Mid-inguinal point

The mid-inguinal point lies at the midpoint of a line between the pubic symphysis and the anterior superior iliac spine. In adults, it is the approximate surface marking of the femoral artery (just below the ligament) and the deep inguinal ring (just above the ligament) (Hale et al 2010).

Inferior epigastric artery

The inferior epigastric artery lies along the medial border of the deep inguinal ring immediately above the inguinal ligament. Up to the level of the umbilicus, it follows a course that lies approximately 40% of the distance between the midline and a sagittal plane running through the anterior superior iliac spine (Epstein et al 2004). The surface anatomy of the artery is particularly important in laparoscopic surgery because trocar insertion may injure the vessel, causing a rectus sheath haematoma or intra-abdominal bleeding. The artery can be avoided if a trocar is inserted at least two-thirds of the distance along a horizontal line between the midline and a sagittal plane passing through the anterior superior iliac spine.

Superficial reflexes

Cremasteric reflex

Stroking the skin of the medial side of the thigh evokes a reflex contraction of cremaster, which elevates the ipsilateral testis. The reflex is mediated by the genitofemoral nerve (L1 and L2 nerve roots), although the afferent arc of the reflex may be via sensory fibres in the ilioinguinal nerve. The reflex is usually absent if there is torsion of the testicle.

Superficial abdominal reflex

Gently stroking each of the four quadrants of the anterior abdominal wall normally elicits a visible contraction of ipsilateral abdominal muscles. The reflex can be used to help localize lesions in the spinal cord but is now of minor clinical significance because of the availability of neuroimaging and the difficulties of eliciting the reflex in multiparous women, the elderly and the obese (Gosavi and Lo 2014).

INTRA-ABDOMINAL VISCERA

The surface markings of the intra-abdominal viscera are variable and depend on age, sex, body habitus, nutritional state, phase of ventilation and body position. The availability of cross-sectional and ultrasound medical imaging of the abdominal viscera has led to a decline in the use of surface anatomy, except for descriptive purposes. The following descriptions are, at best, regarded as the most common or approximate markings in a healthy supine adult (**Fig. 59.11B,C**). The spinous processes of the lumbar vertebrae can be used to locate vertebral levels, with the lower part of the spinous process marking the level of the intervertebral disc below the correspondingly numbered vertebra.

Diaphragm

In the supine position at end-tidal inspiration, the dome of the diaphragm is most frequently located level with the fifth intercostal space in the mid-clavicular line on the right and the sixth rib in the mid-clavicular line on the left, and ranges from the fourth intercostal space to below the costal margin (Mirjalili et al 2012c) (see Fig. 59.11B) (see Ch. 55 for further discussion of the diaphragm). These levels are approximate, not only because of individual variation but also because diaphragmatic excursion in the erect position during quiet breathing is about 2 cm and increases to about 7 cm during deep breathing (Boussuges et al 2009). The dome of the diaphragm can reach as high as the fourth rib on maximal expiration.

Stomach

The gastro-oesophageal junction lies to the left of the midline, posterior to the left seventh costal cartilage, at the level of the eleventh thoracic vertebra (range upper T10 to L1/2), with the level being lower in females and higher in the obese (Mirjalili et al 2012b) (see Fig. 59.11B). The stomach lies in a curve within the left hypochondrium and epigastrium, although, when distended, it may lie as far down as the umbilical or suprapubic regions. The epigastrium is the usual place to auscultate for a 'succussion splash' caused by gastric outlet obstruction or stasis, and is also the region where a thickened pylorus is palpable in infantile hypertrophic pyloric stenosis.

Duodenum

The first part of the duodenum sometimes ascends above the transpyloric plane; the second part usually lies just to the right of the midline, alongside the second and third lumbar vertebrae; the third part usually crosses the midline at the level of the third lumbar vertebra; and the fourth part ascends to the left of the second lumbar vertebra, reaching the transpyloric plane in the region of the lower border of the first lumbar vertebra (see Fig. 59.11B). The duodenojejunal flexure commonly sits at L1 (range lower T11 to upper L3) (Mirjalili et al 2012b).

Small intestine and its mesentery

The small intestine mesentery runs obliquely in a line from a point just to the left of the lower border of the first lumbar vertebra (in the transpyloric plane) towards the right iliac fossa.

Appendix

The appendix is located in the right lower quadrant of the abdomen but is highly variable in its length and position. Studies analysing barium enemas in supine adults have shown the position of appendix base is variably located and rarely precisely at McBurney's point (a point two-thirds of the way along a line joining the anterior superior iliac spine to the umbilicus) (Ramsden et al 1993, Naraynsingh et al 2002).

Liver

At the end of normal tidal inspiration, the inferior border of the liver extends along a line that passes from the right tenth costal cartilage in the mid-axillary line to the left fifth intercostal space/sixth rib in the mid-clavicular line (see Fig. 59.11B). It may be palpable in healthy adults on deep inspiration. The superior border of the liver follows a line that passes from the right fifth rib or intercostal space in the mid-clavicular line to the left fifth intercostal space/sixth rib in the mid-clavicular line, and can be mapped with the contour of the diaphragm (Mirjalili et al 2012c). This border curves slightly downwards at its centre and crosses the midline behind the xiphisternal joint. The right border of the liver is curved to the right and joins the upper and lower right limits. The upper border of the liver may be defined by dullness to percussion when compared with the resonance of the lungs above.

Gallbladder

The fundus of the gallbladder is commonly identified with the tip of the ninth costal cartilage (in the transpyloric plane), near the junction of the linea semilunaris with the costal margin (see Fig. 59.11B). Recent data have shown that the fundus lies in the transpyloric plane in approximately one-third of supine individuals and below the plane in most others (Mirjalili et al 2012b).

Spleen

The spleen is traditionally described as sitting on the left posterolateral abdominal wall deep to ribs 9–11. Recent data show that the spleen sits deep to ribs 10–12 in 50% of subjects and deep to ribs 9–11 in 25% of supine individuals; its long axis corresponds most closely to the eleventh rib or tenth rib, respectively, and passes anterior to the mid-axillary line in most subjects (Mirjalili et al 2012b) (see Fig. 59.11C). The spleen extends from a point about 5 cm to the left of the posterior midline at the level of the eleventh thoracic spine and extends about 3 cm anterior to the mid-axillary line. Its normal size approximates roughly to that of the individual's clenched fist.

RETROPERITONEAL VISCERA

The surface projections of the retroperitoneal viscera are reasonably reliable but have limited clinical utility since most interventions and procedures are guided by medical imaging.

Pancreas

The surface projection of the head of the pancreas lies within the duodenal curve on the right side of the second lumbar vertebra; the neck lies in the transpyloric plane, level with the L1/2 intervertebral disc; and the body passes obliquely up and to the left towards the spleen, lying slightly above the transpyloric plane, near the tail.

Kidney

The right kidney usually lies, on average, 2 cm lower than the left, although, in 10% of cases, the left kidney sits lower than the right (Mirjalili et al 2012b). The vertebral limits of the left kidney are T12–L3 or L4, while those for the right kidney are L1–L4 (overall range upper T11 to lower L5) (see Fig. 59.11C). The upper poles of both kidneys lie anterior to rib 12, and they lie anterior to the rib 11 in 30% (left) and 10% (right) of subjects. In supine adults at end-tidal inspiration, the centre of the renal hilum usually lies at L1/2 or L2 on the left and at a slightly lower vertebral level on the right. It is important to note that both kidneys move vertically by a mean of about 2 cm during deep respiration and both can descend by several centimetres when moving from lying to standing (Schwartz et al 1994, Reiff et al 1999).

The length of the normal adult kidney, measured along its long axis, is approximately 11.5 cm in men and 11.0 cm in women (Hale et al 2010); the left kidney is a few millimetres longer than the right (Cheong et al 2007, Glodny et al 2009, Mirjalili et al 2012b). There is a small reduction in renal length beyond 50 years of age (Glodny et al 2009). Each kidney is approximately 5–6 cm wide. Both its longitudinal and transverse axes are slightly oblique, such that the upper pole of each kidney is nearer the midline and the hilum is more anterior than the lateral surface. The centre of the hilum is approximately 5–6 cm from the midline.

The lower pole of the normal right kidney may occasionally be felt in thin individuals by bimanual palpation via the renal angle on full inspiration.

Ureter

The ureter descends on either side from approximately the level of the transpyloric plane, slightly lower on the right, about 5 cm from the midline. Each passes downwards just medial to the tips of the transverse processes of the lumbar vertebrae. In the pelvic cavity, each ureter curves medially to enter the base of the bladder.

Abdominal aorta and branches

The abdominal aorta begins at the level of the body of the twelfth thoracic vertebra near the midline. It descends and bifurcates at approximately L4 (range lower L3 to lower L5), usually just to the left of the midline and up to 3 cm caudal to the umbilicus (Mirjalili et al 2012b). The position of the aortic bifurcation is shifted slightly proximally with a greater degree of lumbar lordosis (Moussallem et al 2012). The pulsations of the aorta can be felt in a thin, supine individual by pressing firmly in the midline backwards on to the lower lumbar spine. An easily palpable aorta in an obese person should raise the suspicion of an aneurysm.

Unpaired visceral arteries

The coeliac trunk arises from the aorta immediately after it enters the abdomen at T12; the superior mesenteric artery arises most commonly at L1, close to the transpyloric plane; and the inferior mesenteric artery usually arises at L3 (Mirjalili et al 2012b).

Renal arteries

The renal arteries most commonly arise from between lower L1 and upper L2, with the lower border of L1 (approximately, the transpyloric plane) being most common on both sides (Mirjalili et al 2012b).

Iliac arteries

The surface projection of the common iliac artery corresponds to the superior third of a broad line, which is slightly convex laterally, from the aortic bifurcation (see above) to a point midway between the anterior superior iliac spine and the pubic symphysis. The external iliac artery corresponds to the inferior two-thirds of this line.

Inferior vena cava

The inferior vena cava most commonly forms at L5 in the transtubercular plane (range upper L4 to upper S1) approximately to the right of the midline (Mirjalili et al 2012b). The inferior vena cava leaves the abdomen by traversing the diaphragm at the level of the eleventh thoracic vertebra.

PELVIS

The posterior superior iliac spines lie at the level of the second sacral segment (McGaugh et al 2007); some individuals have an overlying sacral dimple. This palpable bony landmark corresponds approximately to the termination of the dural sac (Senoglu et al 2013) and the middle of the sacroiliac joint. The second dorsal sacral foramina lies approximately 2–3 cm medial to the posterior superior iliac spine on each side at an angle of 45° (McGrath and Stringer 2011). The latter is potentially useful when localizing the branches of the dorsal sacral rami in the treatment of refractory sacroiliac pain.

Sciatic nerve

The sciatic nerve leaves the pelvis approximately one-third of the way along a line between the posterior superior iliac spine and the ischial tuberosity and enters the thigh approximately half way between the greater trochanter and ischial tuberosity (see Fig. 78.16) (Currin et al 2014).

Other surface landmarks for pelvic structures are described in Chapter 78.

COMMON CLINICAL PROCEDURES

Pneumoperitoneum for laparoscopy

A pneumoperitoneum is frequently established by accessing the peritoneal cavity just below the umbilicus. An incision through the linea alba allows access to the peritoneum at a point where there is relatively little extraperitoneal fat. Additional working ports are inserted through the anterior abdominal wall, avoiding major vessels such as the inferior epigastric artery.

Surgical incisions

Most surgical incisions are sited according to surgical imperatives rather than anatomical constraints. Access to the peritoneal cavity is gained by dividing or splitting muscles. Common approaches include the midline incision through the relatively avascular linea alba, the transverse suprapubic (Pfannenstiel) incision for pelvic procedures, and, in infants and children, the upper abdominal transverse incision. In recent years, progressively more abdominal surgery has been performed using laparoscopic procedures (minimally invasive or 'keyhole' surgery).

Intestinal stomas (ileostomy, colostomy)

When possible, intestinal stomas are usually formed through transrectus incisions. A cruciate incision is made in the anterior rectus sheath and the muscle fibres are split, avoiding injury to the epigastric vessels. This incision offers the advantage that fibres of rectus abdominis support the stoma, providing a dynamic, contractile surround that tends to reduce the risk of herniation occurring around the stoma.

Suprapubic catheterization

The urinary bladder may be accessed for short- or long-term catheterization through the anterior abdominal wall. As the bladder fills, the upper part of its dome comes to lie in the preperitoneal 'space' in the suprapubic region, where it can be relatively easily accessed by a midline transcutaneous puncture through the linea alba.

Endoscopic surgery

Endoscopic surgery is a broad term that encompasses all types of surgery that are performed using flexible or rigid fibreoptic endoscopes inserted through natural body orifices or small surgical incisions. It overlaps with the field of minimally invasive surgery. The aim is to minimize or eliminate external incisions and hasten the patient's recovery, whilst providing optimum treatment of the pathology with an acceptably low risk of complications. In the abdomen and pelvis, therefore, it includes therapeutic procedures performed using an endoscope inserted via the mouth (gastroscope or duodenoscope), anus (proctoscope, sigmoidoscope or colonoscope), urethra (cystoscope or ureteroscope) or vagina, or via small surgical incisions in the anterior or posterior abdominal wall (laparoscope and other endoscopic systems).

A refinement of the technique is natural orifice transluminal endoscopic surgery, in which abdominal operations are performed using an endoscope inserted through a natural orifice (e.g. mouth, anus, urethra) and then through an incision in the viscus that has been entered (e.g. stomach, bladder, vagina). For example, pelvic and intra-abdominal contents can be accessed through the vagina via the recto-uterine pouch. Hysterectomy, oophorectomy, pelvic organ prolapse repair and incontinence surgery are commonly performed using transvaginal techniques. This approach avoids the morbidity of abdominal incision but there are risks of sciatic, femoral and common fibular nerve injury from prolonged surgery in the lithotomy position.

A further development of endoscopic minimally invasive surgery has been the introduction of robotically assisted surgery, which allows the surgeon to manipulate precision instruments from a console that is remote from the patient whilst viewing the operative field in three dimensions. The procedure is becoming increasingly common in prostatectomy.

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